

Lecture 7

MULTI-STEP-AHEAD FORECASTING

Multi-step-ahead forecasting

Formalities

One-step-ahead forecasting review

MSE-optimal forecast of Y_{n+1} given data $\mathcal{Y}_n$:

$$E(Y_{n+1} \mid \mathcal{Y}_n)$$

One-step-ahead forecast errors:

$$U_{n+1} = Y_{n+1} - E(Y_{n+1} \mid \mathcal{Y}_n)$$

satisfy

$$E(U_{n+1} \mid \mathcal{Y}_n) = 0$$

$\Rightarrow$ errors have no autocorrelation at any lag.

Multi-step-ahead forecasting

MSE-optimal forecast of Y_{n+h} given data $\mathcal{Y}_n$:

$$E(Y_{n+h} \mid \mathcal{Y}_n)$$

Multi-step-ahead forecast errors:

$$U_{n+h \mid n} = Y_{n+h} - E(Y_{n+h} \mid \mathcal{Y}_n)$$

satisfy

$$E(U_{n+h \mid n} \mid \mathcal{Y}_n) = 0$$

$\Rightarrow$ autocorrelation???

Multi-step-ahead forecast errors

For any t : $E(U_{t+h} | t | \mathcal{Y}_t) = 0$

Unconditional mean:

$$E(U_{t+h} | t) = E(E(U_{t+h} | t | \mathcal{Y}_t)) \quad (\text{LIE})$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h} | t | \mathcal{Y}_t) = 0$

Unconditional mean:

$$\begin{aligned} E(U_{t+h} | t) &= E(E(U_{t+h} | t | \mathcal{Y}_t)) && (\text{LIE}) \\ &= 0 \end{aligned}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

Autocovariance at lag $j \geq h$:

$$\begin{aligned} & \text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) \\ &= E(U_{t+h|t} U_{t+h-j|t-j}) \quad \text{since } E(U_{t+h|t}) = 0 \text{ for all } t \end{aligned}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

Autocovariance at lag $j \geq h$:

$$\begin{aligned} & \text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) \\ &= E(U_{t+h|t} U_{t+h-j|t-j}) \quad \text{since } E(U_{t+h|t}) = 0 \text{ for all } t \\ &= E(E(U_{t+h|t} U_{t+h-j|t-j} | \mathcal{Y}_t)) \quad (\text{LIE}) \end{aligned}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

Autocovariance at lag $j \geq h$:

$$\begin{aligned} & \text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) \\ &= E(U_{t+h|t} U_{t+h-j|t-j}) \quad \text{since } E(U_{t+h|t}) = 0 \text{ for all } t \\ &= E(E(U_{t+h|t} U_{t+h-j|t-j} | \mathcal{Y}_t)) \quad (\text{LIE}) \\ &= E(U_{t+h-j|t-j} E(U_{t+h|t} | \mathcal{Y}_t)) \quad \text{since } t+h-j \leq t \end{aligned}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

Autocovariance at lag $j \geq h$:

$$\begin{aligned} & \text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) \\ &= E(U_{t+h|t} U_{t+h-j|t-j}) \quad \text{since } E(U_{t+h|t}) = 0 \text{ for all } t \\ &= E(E(U_{t+h|t} U_{t+h-j|t-j} | \mathcal{Y}_t)) \quad (\text{LIE}) \\ &= E(U_{t+h-j|t-j} E(U_{t+h|t} | \mathcal{Y}_t)) \quad \text{since } t+h-j \leq t \\ &= 0 \end{aligned}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

Autocovariance at lag $j < h$:

$$\begin{aligned} & \text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) \\ &= E(U_{t+h|t} U_{t+h-j|t-j}) \quad \text{since } E(U_{t+h|t}) = 0 \text{ for all } t \\ &= E(E(U_{t+h|t} U_{t+h-j|t-j} | \mathcal{Y}_t)) \quad (\text{LIE}) \\ &\neq E(U_{t+h-j|t-j} E(U_{t+h|t} | \mathcal{Y}_t)) \quad \text{since } t+h-j > t \end{aligned}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} \mid \mathcal{Y}_t) = 0$

$$\text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) = \begin{cases} ? & j = 1, \dots, h-1 \\ 0 & j = h, h+1, \dots \end{cases}$$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

$$\text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) = \begin{cases} ? & j = 1, \dots, h-1 \\ 0 & j = h, h+1, \dots \end{cases}$$

h -step-ahead forecast errors have zero autocorrelation at lags $h, h+1, \dots$

Multi-step-ahead forecast errors

For any t : $E(U_{t+h|t} | \mathcal{Y}_t) = 0$

$$\text{cov}(U_{t+h|t}, U_{t+h-j|t-j}) = \begin{cases} ? & j = 1, \dots, h-1 \\ 0 & j = h, h+1, \dots \end{cases}$$

h -step-ahead forecast errors *may* have non-zero autocorrelation at lags $1, \dots, h-1$.

Multi-step-ahead forecasting

Approach 1

Multi-step forecasts from 1-step models

The most common approach:

- specify model of $E(Y_t | \mathcal{Y}_{t-1})$, eg ARIMA
- use “extended” LIE

$$E(Y | \mathcal{F}) = E(E(Y | \mathcal{G}) | \mathcal{F}) \quad \text{if } \mathcal{F} \subset \mathcal{G}$$

Multi-step forecasts from 1-step models

The most common approach:

- specify model of $E(Y_t | \mathcal{Y}_{t-1})$, eg ARIMA
- use “extended” LIE

$$E(Y | \mathcal{F}) = E(E(Y | \mathcal{G}) | \mathcal{F}) \quad \text{if } \mathcal{F} \subset \mathcal{G}$$

to *derive* multi-step-ahead forecasts. Eg:

$$E(Y_{n+2} | \mathcal{Y}_n) = E(E(Y_{n+2} | \mathcal{Y}_{n+1}) | \mathcal{Y}_n)$$

since $\mathcal{Y}_n \subset \mathcal{Y}_{n+1}$

Multi-step forecasts from 1-step models

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to *derive* multi-step-ahead forecasts. Eg:

$$E(Y_{n+2} | \mathcal{Y}_n) = E(E(Y_{n+2} | \mathcal{Y}_{n+1}) | \mathcal{Y}_n)$$

since $\mathcal{Y}_n \subset \mathcal{Y}_{n+1}$

↑

More data at time $n + 1$ than time n .

Multi-step forecasts from 1-step models

The most common approach:

- specify model of $E(Y_t | \mathcal{Y}_{t-1})$, eg ARIMA
- use “extended” LIE

$$E(Y | \mathcal{F}) = E(E(Y | \mathcal{G}) | \mathcal{F}) \quad \text{if } \mathcal{F} \subset \mathcal{G}$$

to *derive* multi-step-ahead forecasts. Eg:

$$E(Y_{n+2} | \mathcal{Y}_n) = E(E(Y_{n+2} | \mathcal{Y}_{n+1}) | \mathcal{Y}_n)$$



1-step-ahead forecast

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

2-step-ahead forecast:

$$E(Y_{n+2} \mid \mathcal{Y}_n) = E(E(Y_{n+2} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n)$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

2-step-ahead forecast:

$$\begin{aligned} E(Y_{n+2} \mid \mathcal{Y}_n) &= E(E(Y_{n+2} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1 Y_{n+1} \mid \mathcal{Y}_n) \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

2-step-ahead forecast:

$$\begin{aligned} E(Y_{n+2} \mid \mathcal{Y}_n) &= E(E(Y_{n+2} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1 Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1 E(Y_{n+1} \mid \mathcal{Y}_n) \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

2-step-ahead forecast:

$$\begin{aligned} E(Y_{n+2} \mid \mathcal{Y}_n) &= E(E(Y_{n+2} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1 Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1 E(Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1 \phi_1 Y_n \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

2-step-ahead forecast:

$$\begin{aligned} E(Y_{n+2} \mid \mathcal{Y}_n) &= E(E(Y_{n+2} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1 Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1 E(Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1 \phi_1 Y_n \\ &= \phi_1^2 Y_n \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

$$E(Y_{n+2} \mid \mathcal{Y}_n) = \phi_1^2 Y_n$$

3-step-ahead forecast:

$$E(Y_{n+3} \mid \mathcal{Y}_n) = E(E(Y_{n+3} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n)$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

$$E(Y_{n+2} \mid \mathcal{Y}_n) = \phi_1^2 Y_n$$

3-step-ahead forecast:

$$\begin{aligned} E(Y_{n+3} \mid \mathcal{Y}_n) &= E(E(Y_{n+3} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1^2 Y_{n+1} \mid \mathcal{Y}_n) \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

$$E(Y_{n+2} \mid \mathcal{Y}_n) = \phi_1^2 Y_n$$

3-step-ahead forecast:

$$\begin{aligned} E(Y_{n+3} \mid \mathcal{Y}_n) &= E(E(Y_{n+3} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1^2 Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1^2 E(Y_{n+1} \mid \mathcal{Y}_n) \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

$$E(Y_{n+2} \mid \mathcal{Y}_n) = \phi_1^2 Y_n$$

3-step-ahead forecast:

$$\begin{aligned} E(Y_{n+3} \mid \mathcal{Y}_n) &= E(E(Y_{n+3} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1^2 Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1^2 E(Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1^2 \phi_1 Y_n \end{aligned}$$

Example. AR(1) model

$$E(Y_{n+1} \mid \mathcal{Y}_n) = \phi_1 Y_n$$

$$E(Y_{n+2} \mid \mathcal{Y}_n) = \phi_1^2 Y_n$$

3-step-ahead forecast:

$$\begin{aligned} E(Y_{n+3} \mid \mathcal{Y}_n) &= E(E(Y_{n+3} \mid \mathcal{Y}_{n+1}) \mid \mathcal{Y}_n) \\ &= E(\phi_1^2 Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1^2 E(Y_{n+1} \mid \mathcal{Y}_n) \\ &= \phi_1^3 Y_n \end{aligned}$$

Example. AR(1) model

For any $h = 1, 2, \dots$

$$E(Y_{n+h} \mid \mathcal{Y}_n) = \phi_1^h Y_n$$

This process can be applied to derive the h -step-ahead forecasting formulae for any ARIMA model.

(But R **forecast** handles these formulae for us!)

Short- vs long-term forecasting

What happens as h increases?

Consider the AR(1) model

$$E(Y_t \mid \mathcal{Y}_{t-1}) = \phi_1 Y_{t-1}$$

with h -step-ahead forecast function

$$E(Y_{n+h} \mid \mathcal{Y}_n) = \phi_1^h Y_n$$

What happens as h increases?

Consider the AR(1) model

$$E(Y_t \mid \mathcal{Y}_{t-1}) = \phi_1 Y_{t-1}$$

with h -step-ahead forecast function

$$E(Y_{n+h} \mid \mathcal{Y}_n) = \phi_1^h Y_n$$

As $h \rightarrow \infty$:

- $|\phi_1| < 1 \quad \Rightarrow \quad \phi_1^h \rightarrow 0$
- $\phi_1 = 1 \quad \Rightarrow \quad \phi_1^h = 1$ for every h
- $|\phi_1| > 1 \quad \Rightarrow \quad \phi_1^h \rightarrow \infty$

What happens as h increases?

Consider the AR(1) model

$$E(Y_t \mid \mathcal{Y}_{t-1}) = \phi_1 Y_{t-1}$$

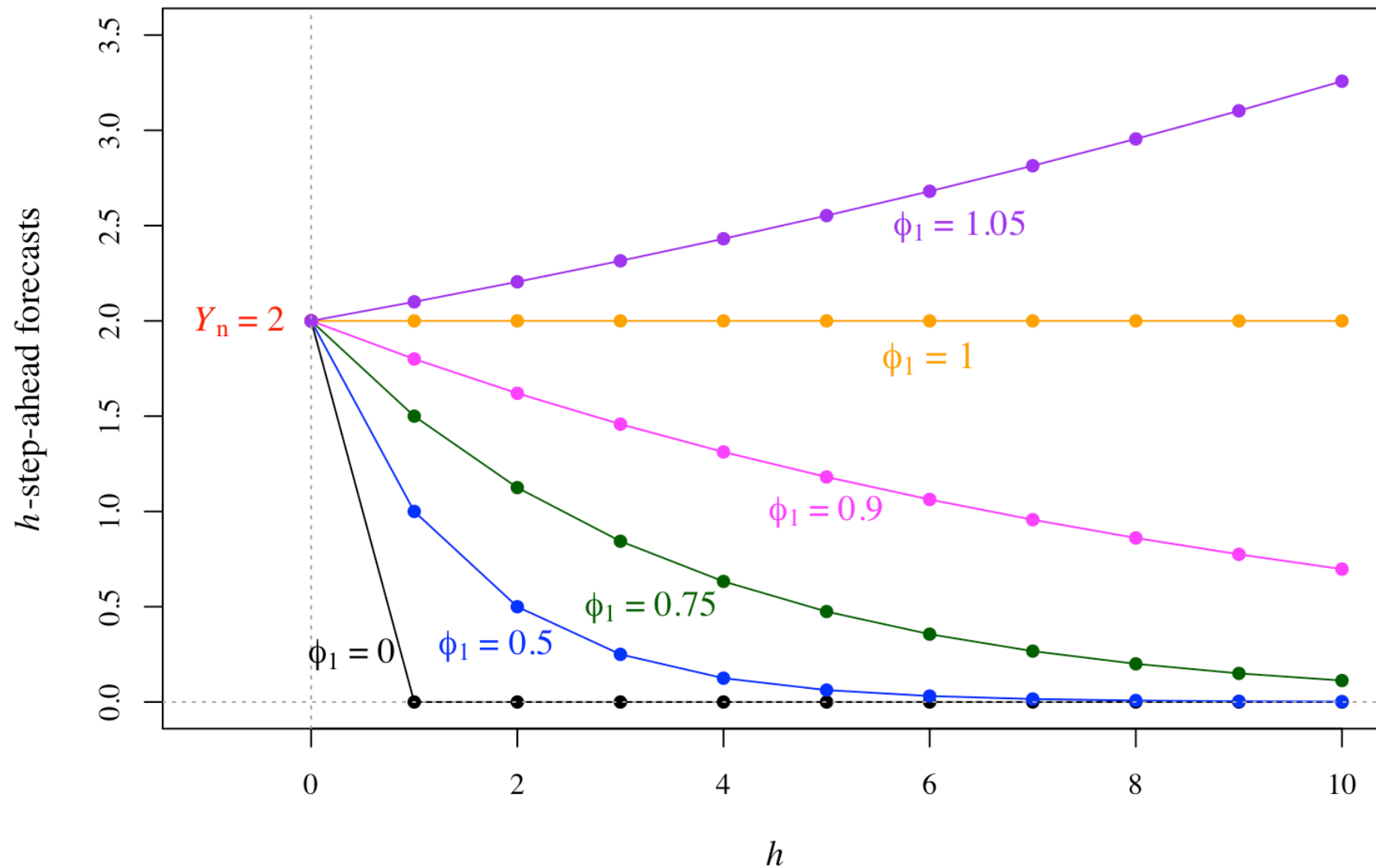
with h -step-ahead forecast function

$$E(Y_{n+h} \mid \mathcal{Y}_n) = \phi_1^h Y_n$$

As $h \rightarrow \infty$:

- $|\phi_1| < 1 \quad \Rightarrow \quad E(Y_{n+h} \mid \mathcal{Y}_n) \rightarrow 0$
- $\phi_1 = 1 \quad \Rightarrow \quad E(Y_{n+h} \mid \mathcal{Y}_n) = Y_n$ for every h
- $|\phi_1| > 1 \quad \Rightarrow \quad E(Y_{n+h} \mid \mathcal{Y}_n) \rightarrow \pm\infty$

Numerical illustration with $Y_n = 2$



Forecasting with a trend function

Suppose X_t specifies a *deterministic* trend:

$$Y_t = X_t' \beta + Z_t$$

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Forecasting with a trend function

Suppose X_t specifies a *deterministic* trend:

$$Y_t = X_t' \beta + Z_t$$

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Then

$$E(Y_{n+h} \mid \mathcal{Y}_n) = X_{n+h}' \beta + E(Z_{n+h} \mid \mathcal{Y}_n)$$

↑

Future values of X_t are *known*.

i.e. no need to forecast these

Forecasting with a trend function

Suppose X_t specifies a *deterministic* trend:

$$Y_t = X_t' \beta + Z_t$$

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Then

$$\begin{aligned} E(Y_{n+h} \mid \mathcal{Y}_n) &= X_{n+h}' \beta + E(Z_{n+h} \mid \mathcal{Y}_n) \\ &= X_{n+h}' \beta + E(Z_{n+h} \mid \mathcal{Z}_n) \end{aligned}$$

X_t is deterministic $\Rightarrow \mathcal{Y}_n$ and $\mathcal{Z}_n$ are equivalent.

Forecasting with a trend function

Suppose X_t specifies a *deterministic* trend:

$$Y_t = X_t' \beta + Z_t$$

$$E(Z_t | \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Then

$$\begin{aligned} E(Y_{n+h} | \mathcal{Y}_n) &= X_{n+h}' \beta + E(Z_{n+h} | \mathcal{Y}_n) \\ &= X_{n+h}' \beta + E(Z_{n+h} | \mathcal{Z}_n) \\ &= X_{n+h}' \beta + \phi_1^h Z_n \end{aligned}$$

Forecasting with a trend function

Suppose X_t specifies a *deterministic* trend:

$$Y_t = X_t' \beta + Z_t$$

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Then

$$\begin{aligned} E(Y_{n+h} \mid \mathcal{Y}_n) &= X_{n+h}' \beta + E(Z_{n+h} \mid \mathcal{Y}_n) \\ &= X_{n+h}' \beta + E(Z_{n+h} \mid \mathcal{Z}_n) \\ &= X_{n+h}' \beta + \phi_1^h Z_n \\ &= X_{n+h}' \beta + \phi_1^h (Y_n - X_n' \beta) \end{aligned}$$

Forecasting with a trend function

Suppose X_t specifies a *deterministic* trend:

$$Y_t = X'_t \beta + Z_t$$

$$E(Z_t | \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Then

$$E(Y_{n+h} | \mathcal{Y}_n) = X'_{n+h} \beta + \phi_1^h (Y_n - X'_n \beta)$$

forecast for

time $n + h$

trend at

time $n + h$

forecast for

deviation from

trend at time $n + h$

Forecasting with a trend function

$$E(Y_{n+h} | \mathcal{Y}_n) = X'_{n+h}\beta + \phi_1^h (Y_n - X'_n\beta)$$

As $h \rightarrow \infty$:

$$|\phi_1| < 1 \quad \Rightarrow \quad E(Y_{n+h} | \mathcal{Y}_n) - X'_{n+h}\beta \rightarrow 0$$

Stationarity implies the forecasts converge to the trend function as the forecast horizon h increases.

Forecasting with a trend function

$$E(Y_{n+h} | \mathcal{Y}_n) = X'_{n+h}\beta + \phi_1^h (Y_n - X'_n\beta)$$

As $h \rightarrow \infty$:

$$\phi_1 = 1 \Rightarrow E(Y_{n+h} | \mathcal{Y}_n) - X'_{n+h}\beta = Y_n - X'_n\beta$$

A **unit root** implies the time series is **not forecast** to return to its **trend**, regardless of length of time horizon.

Forecasting with a trend function

$$E(Y_{n+h} | \mathcal{Y}_n) = X'_{n+h}\beta + \phi_1^h (Y_n - X'_n\beta)$$

As $h \rightarrow \infty$:

$$|\phi_1| > 1 \quad \Rightarrow \quad E(Y_{n+h} | \mathcal{Y}_n) - X'_{n+h}\beta \rightarrow \pm\infty$$

An **explosive** AR model implies the **forecasts** diverge from the **trend function** as the forecast horizon h increases.

Forecasting with a trend function + ARMA

$$Y_t = X_t' \beta + Z_t$$

$$Z_t = \phi_1 Z_{t-1} + \dots + \phi_p Z_{t-p} \\ + U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q}$$

- Z_t stationary $\Rightarrow$ as $h \rightarrow \infty$:
 $E(Y_{t+h} | \mathcal{Y}_n) - X_{n+h}' \beta \rightarrow 0$
- Y_t is called “*trend stationary*” or “*mean reverting*” or “*mixingale*”.

Forecasting with a trend function + ARMA

$$Y_t = X_t' \beta + Z_t$$

$$Z_t = \phi_1 Z_{t-1} + \dots + \phi_p Z_{t-p} \\ + U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q}$$

- Z_t unit root $\Rightarrow$ as $h \rightarrow \infty$:

$$E(Y_{t+h} | \mathcal{Y}_n) - X_{n+h}' \beta \rightarrow Y_n - X_n' \beta$$

- Z_t is called “*difference stationary*”.

Multi-step-ahead forecasting

Approach 2

Direct forecasting

Usual approach:

- model $E(Y_{n+1}|\mathcal{F}_n)$
- then derive $E(Y_{n+h}|\mathcal{F}_n)$ for $h = 1, 2, \dots$

“Direct forecasting”:

- model $E(Y_{n+h}|\mathcal{F}_n)$ for $h = 1, 2, \dots$

Direct forecasting with AR models

$$E(Y_t | \mathcal{Y}_{t-1}) = \phi_1^{(1)} Y_{t-1} + \phi_2^{(1)} Y_{t-2} + \dots$$

$$E(Y_t | \mathcal{Y}_{t-2}) = \phi_2^{(2)} Y_{t-2} + \phi_3^{(2)} Y_{t-3} + \dots$$

$$E(Y_t | \mathcal{Y}_{t-3}) = \phi_3^{(3)} Y_{t-3} + \phi_4^{(3)} Y_{t-4} + \dots$$

$\vdots$

i.e. specify and estimate a *separate* model for each forecast horizon h

Direct forecasting with AR models

$$E(Y_t | \mathcal{Y}_{t-1}) = \phi_1^{(1)} Y_{t-1} + \phi_2^{(1)} Y_{t-2} + \dots$$

$$E(Y_t | \mathcal{Y}_{t-2}) = \phi_2^{(2)} Y_{t-2} + \phi_3^{(2)} Y_{t-3} + \dots$$

$$E(Y_t | \mathcal{Y}_{t-3}) = \phi_3^{(3)} Y_{t-3} + \phi_4^{(3)} Y_{t-4} + \dots$$

⋮

The *first* AR lag in each model is $h = 1, 2, 3, \dots$

Direct forecasting with AR models

$$E(Y_t | \mathcal{Y}_{t-1}) = \phi_1^{(1)} Y_{t-1} + \dots + \phi_{\mathbf{p}^{(1)}}^{(1)} Y_{t-\mathbf{p}^{(1)}}$$

$$E(Y_t | \mathcal{Y}_{t-2}) = \phi_2^{(2)} Y_{t-2} + \dots + \phi_{\mathbf{p}^{(2)}}^{(2)} Y_{t-\mathbf{p}^{(2)}}$$

$$E(Y_t | \mathcal{Y}_{t-3}) = \phi_3^{(3)} Y_{t-3} + \dots + \phi_{\mathbf{p}^{(3)}}^{(3)} Y_{t-\mathbf{p}^{(3)}}$$

⋮

AR lag order may differ across $h = \mathbf{1}, \mathbf{2}, \mathbf{3}, \dots$

Choice of AR lag orders

For each $h = 1, 2, \dots$

- Check residual autocorrelation in models

$$Y_t = \phi_h^{(h)} Y_{t-h} + \dots + \phi_{p^{(h)}}^{(h)} Y_{t-p^{(h)}} + U_t^{(h)}$$

for $p^{(h)} = h, \dots, p_{\max}$.

- Choose $p^{(h)}$ to minimise AIC_c among models that pass the autocorrelation test.

Ljung-Box test for direct AR models

Correct specification requires $U_t^{(h)}$ in

$$Y_t = \phi_h^{(h)} Y_{t-h} + \dots + \phi_{p^{(h)}}^{(h)} Y_{t-p^{(h)}} + U_t^{(h)}$$

must satisfy

$$\text{cov}(U_t^{(h)}, U_{t-j}^{(h)}) = 0 \text{ for } j = h, h+1, \dots$$

Autocovariances at $j = 1, \dots, h$ are unrestricted.

Ljung-Box test for direct AR(p) models

Define $\rho_j^{(h)} = \text{cor}(U_t^{(h)}, U_{t-j}^{(h)})$.

Standard Ljung-Box test hypotheses:

$$H_0 : \rho_1^{(h)} = \dots = \rho_l^{(h)}$$

$$H_1 : \text{at least one of } \rho_1^{(h)}, \dots, \rho_l^{(h)} \neq 0$$

and statistic:

$$LB_l = n(n+2) \sum_{j=1}^l \frac{\hat{\rho}_j^2}{n-j}$$

Ljung-Box test for **direct AR(p)** models

Define $\rho_j^{(h)} = \text{cor}(U_t^{(h)}, U_{t-j}^{(h)})$.

We want Ljung-Box test hypotheses:

$$H_0 : \rho_{\mathbf{h}}^{(h)} = \dots = \rho_l^{(h)}$$

$$H_1 : \text{at least one of } \rho_{\mathbf{h}}^{(h)}, \dots, \rho_l^{(h)} \neq 0$$

and statistic:

$$LB_l^{(h)} = n(n+2) \sum_{j=\mathbf{h}}^l \frac{\hat{\rho}_j^2}{n-j}$$

Ljung-Box test for **direct AR(p)** models

Define $\rho_j^{(h)} = \text{cor}(U_t^{(h)}, U_{t-j}^{(h)})$.

We want Ljung-Box test hypotheses:

$$H_0 : \rho_{\mathbf{h}}^{(h)} = \dots = \rho_l^{(h)}$$

$$H_1 : \text{at least one of } \rho_{\mathbf{h}}^{(h)}, \dots, \rho_l^{(h)} \neq 0$$

and statistic:

$$LB_l^{(h)} = LB_l - LB_{h-1}$$

Ljung-Box test for direct AR(p) models

Define $\rho_j^{(h)} = \text{cor}(U_t^{(h)}, U_{t-j}^{(h)})$.

We want Ljung-Box test hypotheses:

$$H_0 : \rho_{\textcolor{brown}{h}}^{(h)} = \dots = \rho_{\textcolor{violet}{l}}^{(h)}$$

$$H_1 : \text{at least one of } \rho_{\textcolor{brown}{h}}^{(h)}, \dots, \rho_{\textcolor{violet}{l}}^{(h)} \neq 0$$

and statistic:

$$LB_{\textcolor{violet}{l}}^{(\textcolor{brown}{h})} = LB_{\textcolor{violet}{l}} - LB_{\textcolor{violet}{h}-1} \sim \chi_{\textcolor{violet}{l}-\textcolor{green}{p}}^2 \text{ under } H_0$$

Multi-step forecasting summary

Approach 1: “Recursive” forecasting

Specify a **single model** for one-step-ahead forecasting:

$$E(Y_t \mid \mathcal{Y}_{t-1})$$

and *derive/compute* forecasts from **this model**:

$$E(Y_{n+h} \mid \mathcal{Y}_n), \quad h = 1, 2, \dots$$

Approach 2: “Direct” forecasting

Specify **separate models** for h -step-ahead forecasting:

$$E(Y_t | \mathcal{Y}_{t-1}), E(Y_t | \mathcal{Y}_{t-2}), E(Y_t | \mathcal{Y}_{t-3}), \dots$$

Residual autocorrelation checking in $E(Y_t | \mathcal{Y}_{t-h})$:

$$LB_l^{(h)} = LB_l - LB_{h-1}$$

where LB_l is the usual Ljung-Box statistic.

Approach 2: “Direct” forecasting

Specify **separate models** for h -step-ahead forecasting:

$$E(Y_t | \mathcal{Y}_{t-1}), E(Y_t | \mathcal{Y}_{t-2}), E(Y_t | \mathcal{Y}_{t-3}), \dots$$

Residual autocorrelation checking in $E(Y_t | \mathcal{Y}_{t-h})$:

$$LB_l^{(h)} = LB_l - LB_{h-1}$$

Critical / p -values from χ_{l-p}^2 distribution.

Eg. $l = p + 4$ provides 4 d.f. in the test.

Some comparisons

Recursive forecasting:

- requires only one model search
- forecasts are “model consistent”

Direct forecasting:

- does not require derivations of multi-step formulae
- perceived to be more flexible across h